

Supporting Information

Optical Thermometry in Monolayer WS₂: Raman and Photoluminescence Spectroscopy from 5 K to 1273 K

G. Lee¹, A. Borel¹, T. Taniguchi², K. Watanabe³, F. Sirotti¹, and F. Cadiz¹

¹Laboratoire de Physique de la Matière Condensée, CNRS, Ecole polytechnique, Institut Polytechnique de Paris, 91120 Palaiseau, France

²International Center for Materials Nanoarchitectonics, National Institute for Materials Science, 1-1 Namiki, Tsukuba 305-0044, Japan

³Research Center for Functional Materials, National Institute for Materials Science, 1-1 Namiki, Tsukuba 305-0044, Japan

*Email: fabian.cadiz@polytechnique.edu

S1. Raman: Individual (E_{2g}^1 , 2LA) versus Merged-Peak Fitting

In addition to the $A_{1g}(\Gamma)$ mode used as the primary thermometer in the main text, the $E_{2g}^1(\Gamma)$ and 2LA(M) modes (~ 351 and ~ 347 cm^{-1} at room temperature, respectively) were tracked as a function of temperature for both samples. These two modes sit only ~ 5 cm^{-1} apart at room temperature and broaden and redshift with increasing temperature, so that they progressively overlap and become difficult to resolve individually. Each dataset was fitted **in two ways**: (i) as two independent Lorentzian peaks (individual fit), and (ii) as a single Lorentzian standing in for the unresolved pair (merged fit).

Non-encapsulated sample. Figure S1(a,b) shows the individually-fitted E_{2g}^1 and 2LA positions versus temperature. Both peaks are well resolved and the individual fits are stable across the full range; linear fits restricted to $T \geq 600$ K give slopes of -0.0174 cm^{-1}/K (E_{2g}^1) and -0.0137 cm^{-1}/K (2LA). Figure S1(c) shows the merged-peak fit, which is flatter below ~ 500 K and only becomes clearly linear at higher temperature; a linear fit restricted to $T \geq 500$ K yields -0.0179 ± 0.0002 cm^{-1}/K ($R^2 = 0.998$, $n = 15$) – consistent with the individual-peak slopes, since the non-encapsulated sample’s individual fits are already well-behaved.

hBN-encapsulated sample. Figure S1(d,e) shows that the individual fit is far less reliable for this sample: above ~ 900 K, the optimizer is prone to bound-splitting artifacts and chain-drift, and the E_{2g}^1 position visibly deviates from a physical monotonic redshift (it even turns upward in Fig. S1(d)), with error bars several times larger than at low temperature. This motivates treating E_{2g}^1 and 2LA as a single merged Lorentzian peak for this sample (Figure S1(f)), which gives a more stable, if less precise, position estimate. A pooled linear fit over **$T \geq 250$ K, below which the merged position does not decrease with temperature** (combining all four measurement stages: LT, cooldown, 1st heat-up, 2nd heat-up) yields a slope of **-0.0114 ± 0.0007 cm^{-1}/K ($R^2 = 0.915$, $n = 24$)**, consistent in sign and order of magnitude with the non-encapsulated result.

This fit should not, however, be read as a quantitative calibration for the encapsulated sample. Three features of the data argue against it. First, the intensity redistribution described in the main text removes the doublet entirely between ~ 373 K and ~ 923 K, so the fit spans a 550 K gap and is effectively constrained by two separated clusters of points rather than by a continuous trend. Second, below ~ 400 K the merged position is not monotonic: it rises from 349.1 to 350.9 cm^{-1} between 5 K and 293 K before decreasing again. Third, repeated acquisitions at the same nominal temperature differ by up to 2.3 cm^{-1} at high temperature,

several times the corresponding fit uncertainty ($\sim 0.4 \text{ cm}^{-1}$), indicating that the merged position is not reproducible at that level.

The $E_{2g}^1 + 2LA$ doublet is therefore used only as a qualitative, sign-and-order-of-magnitude check. For the non-encapsulated sample it provides a usable secondary indicator above $\sim 500 \text{ K}$; for the encapsulated sample no quantitative slope is claimed. In neither case is it proposed as a thermometer in place of the A_{1g} mode used in the main text.

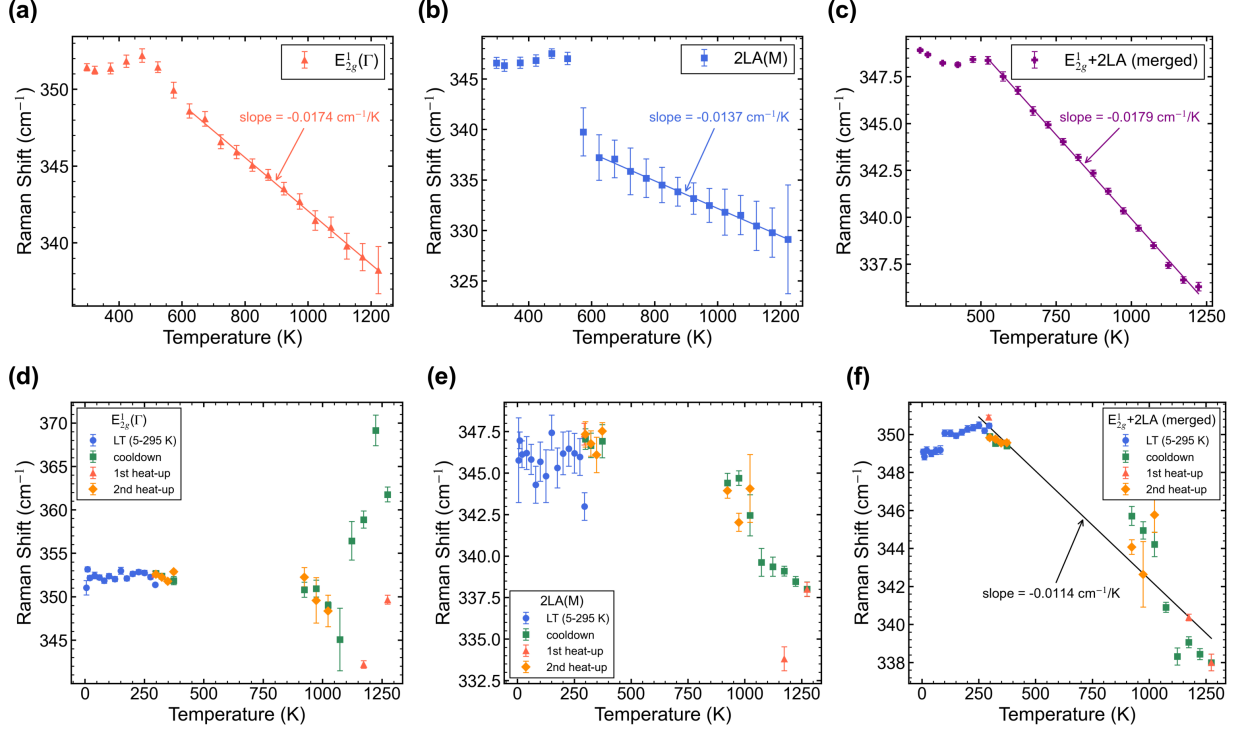


Figure S1: Individual versus merged-peak fitting of the $E_{2g}^1 + 2LA$ doublet. Top row: non-encapsulated sample – (a) E_{2g}^1 individual fit, (b) $2LA$ individual fit, (c) merged-peak fit ($T \geq 500 \text{ K}$ linear fit shown). Bottom row: hBN-encapsulated sample – (d) E_{2g}^1 individual fit, (e) $2LA$ individual fit, both showing instability above $\sim 900 \text{ K}$, (f) merged-peak fit, pooled linear fit over $T \geq 250 \text{ K}$ across all four measurement stages.

S2. Temperature-Dependent Raman Spectra of hBN-Encapsulated WS₂

Figure S2 shows the Raman spectra of the hBN-encapsulated WS₂ monolayer over the full measured temperature range, combining the low-temperature (5 K to 295 K) and high-temperature (RT to 1273 K) datasets. Data points where the peak position could not be reliably extracted are excluded from the analysis. The A_{1g} positions from the cooldown and the two heat-up stages scatter around the same line (Figure S2(b)), with stage-to-stage differences of up to $\sim 0.8 \text{ cm}^{-1}$ but no consistent offset between heating and cooling.

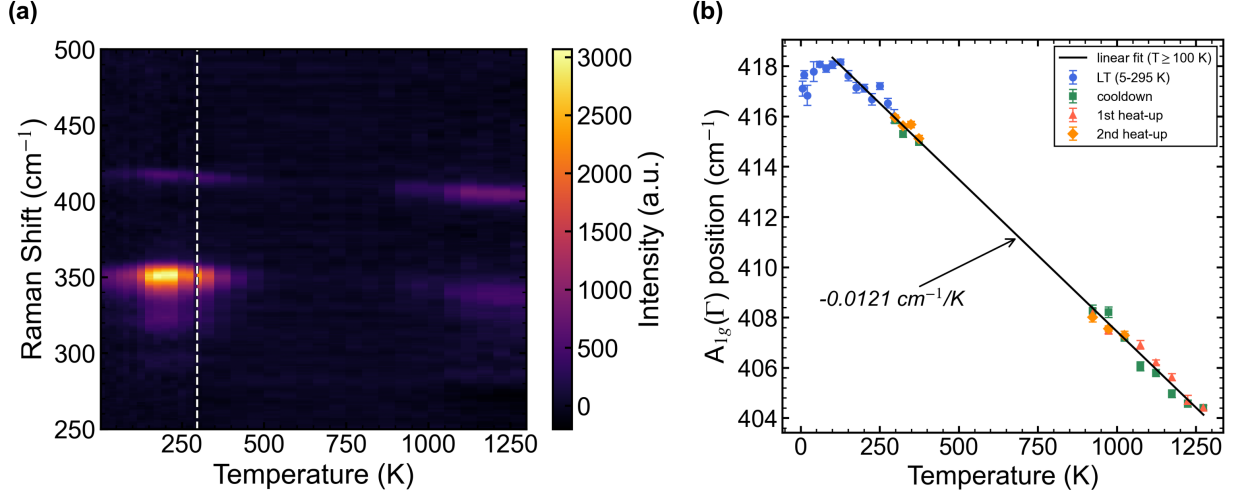


Figure S2: Temperature-dependent Raman response of the hBN-encapsulated WS₂ monolayer. (a) Two-dimensional map of Raman intensity versus temperature (5–1273 K) and Raman shift; the dashed line marks the boundary between the cryostat (5 K–RT) and micro-heater (RT–1273 K) stages. (b) A_{1g} peak position versus temperature for each measurement stage. The solid line is a linear fit to this sample over $T \geq 100$ K ($-0.0121 \text{ cm}^{-1}/\text{K}$), as used in the main text; cooldown and heat-up data scatter around the same line.

S3. Comparison of Bandgap Temperature Models

Three empirical and semi-empirical models for the temperature dependence of the bandgap energy were fitted to the neutral-exciton peak energy $E_0(T)$ extracted from Voigt fits of the PL spectra. The main text uses the Pässler model (Eq. 2); the two alternative models below are compared against it here.

Varshni model (1967) [1]:

$$E(T) = E_0 - \frac{\alpha T^2}{T + \beta}, \quad (1)$$

where E_0 is the zero-temperature bandgap, α is the electron–phonon coupling constant (eV/K), and β is an effective Debye temperature (K). This model is purely empirical and does not reduce to the correct zero-temperature curvature.

O’Donnell–Chen model (1991) [2]:

$$E(T) = E_0 - S\hbar\omega \left[\coth\left(\frac{\hbar\omega}{2k_B T}\right) - 1 \right], \quad (2)$$

where S is a dimensionless Huang–Rhys-like electron–phonon coupling strength and $\hbar\omega$ is the mean phonon energy. At high temperature this model is linear in T , with slope $2Sk_B$.

Figure S3 shows all three model fits (Pässler included for reference) to the data together with the corresponding residuals. The fitted parameters and goodness-of-fit metrics are summarized in Table S1. The Varshni model systematically overestimates the energy at low temperature, while both the Pässler and O’Donnell–Chen models – which share the same physically-motivated saturation to a temperature-independent phonon population at low T – reproduce the data to within the experimental uncertainty over the full range.

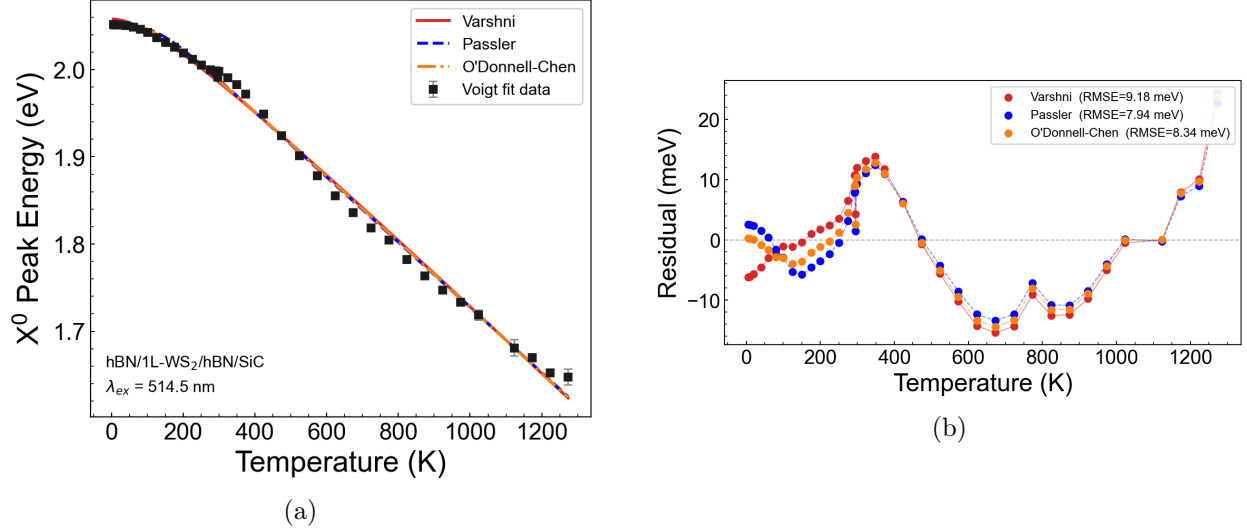


Figure S3: Comparison of bandgap temperature models fitted to the neutral-exciton peak energy of hBN-encapsulated WS₂. (a) Experimental $E_0(T)$ (symbols) and fits using the Varshni (dashed red), Pässler (solid blue), and O'Donnell-Chen (dash-dot orange) models. (b) Fit residuals (data – model) in meV for each model, showing that the Varshni model systematically overestimates the energy at low temperature, while both the Pässler and O'Donnell-Chen models fit the data to within the experimental uncertainty over the full range.

Table S1: Fitted parameters and goodness-of-fit metrics for the three bandgap temperature models (weighted least-squares fit to the neutral-exciton X_A^0 peak energy).

	Varshni	Pässler	O'Donnell-Chen
$E_0(0)$ (eV)	2.0578 ± 0.0037	2.0491 ± 0.0036	2.0514 ± 0.0034
α (10^{-4} eV/K)	3.908 ± 0.116	3.741 ± 0.069	—
β (K)	185.2 ± 41.4	—	—
Θ (K)	—	280.1 ± 28.8	—
p	—	4.04 ± 2.23	—
S	—	—	2.210 ± 0.038
$\hbar\omega$ (meV)	—	—	27.3 ± 3.4
R^2	0.9952	0.9964	0.9961
RMSE (meV)	9.18	7.94	8.34

S4. Voigt Fitting Procedure and Quality Assessment

At each temperature, the X_A^0 emission was fitted with a Voigt profile, the convolution of a Gaussian of standard deviation σ and a Lorentzian of half-width γ , evaluated through the Faddeeva function $w(z)$:

$$V(E) = A \frac{\text{Re}[w(z)]}{\sigma\sqrt{2\pi}}, \quad z = \frac{E - E_0 + i\gamma}{\sigma\sqrt{2}}, \quad (3)$$

plus a constant background B . Below 125 K, where X_A^0 is spectrally isolated, a single Voigt profile was fitted within a ± 22 meV window centred on the peak position obtained at the previous temperature. From 125 K upwards, X_A^0 broadens into the lower-energy trion (X^-) emission, and the full measured spectral window was fitted with two Voigt profiles (X_A^0 and X^-) sharing a common background, with the X^- amplitude constrained not to exceed that of X_A^0 . The Gaussian and Lorentzian widths are $\Gamma_G = 2\sqrt{2\ln 2}\sigma$

and $\Gamma_L = 2\gamma$; the total FWHM Γ_V of the X_A^0 component was obtained numerically from the half-maximum crossings of the fitted profile rather than from an approximate closed-form expression.

Figure S4 shows representative fits at twelve temperatures spanning the full range.

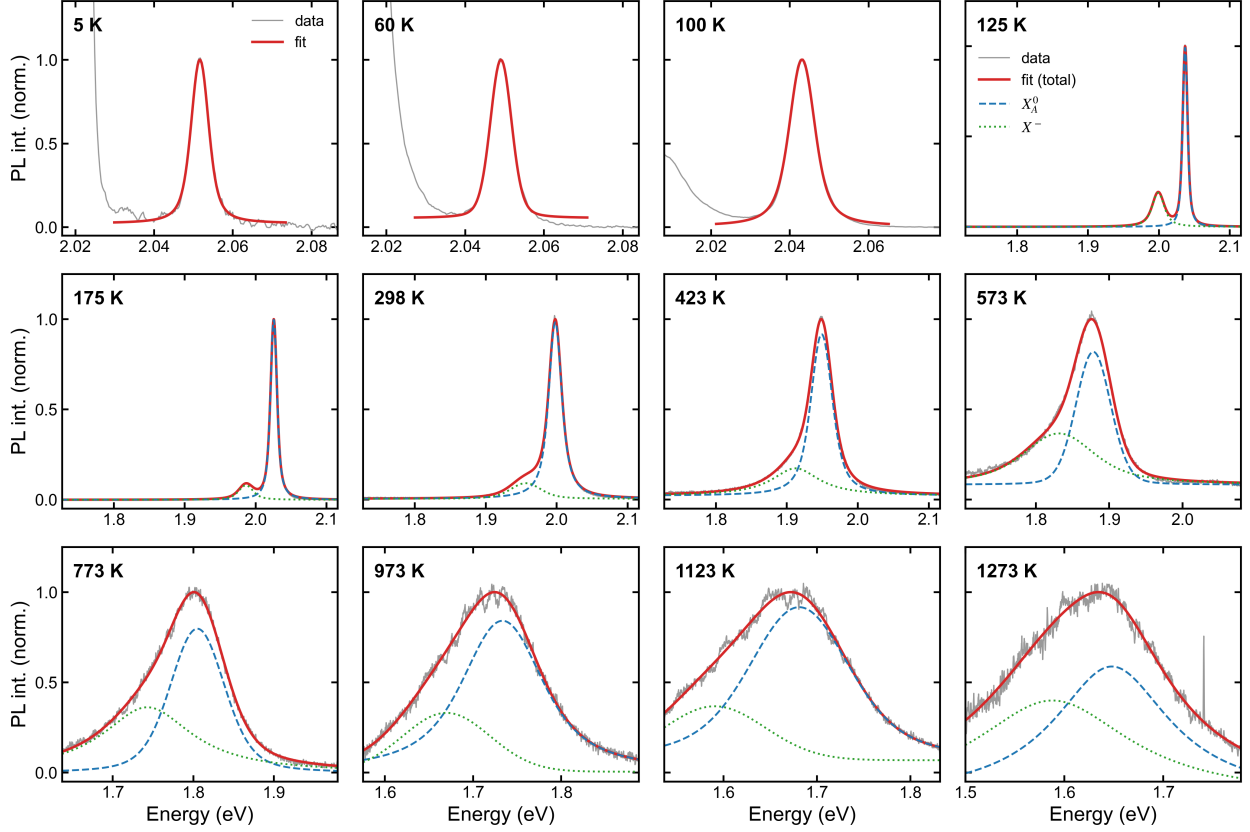


Figure S4: Representative Voigt fits to the PL spectra of hBN-encapsulated WS_2 at twelve temperatures. Grey: data; red: fit. Below 125 K (5, 60 and 100 K) a single Voigt profile is fitted within ± 22 meV of X_A^0 , and only this window is shown. From 125 K upwards the fit (red) is the sum of the X_A^0 (blue dashed) and X^- (green dotted) components over the full acquisition window. Each panel is normalised to the maximum of the fit. At the highest temperatures the low-energy tail of the emission extends beyond the acquisition window.

Note: the Gaussian and Lorentzian widths are strongly correlated in the fit, and at several temperatures the optimizer converges to a bound, so that the individual components are not reliably determined. Only the total Voigt width is reported (Sec. S5). The peak position E_0 is robust to this degeneracy and all points are retained in the $E_0(T)$ dataset.

S5. Photoluminescence Linewidth and Integrated Intensity

Figure S5(a) shows the total Voigt FWHM Γ_V of the X_A^0 emission as a function of temperature. The linewidth increases monotonically from ~ 5.3 meV at 5 K to ~ 140 meV at 1273 K, consistent with an increasing phonon-scattering-limited dephasing rate. Over 5–1000 K the linewidth is well described by the phonon-scattering model $\Gamma(T) = \Gamma_0 + c_1 T + c_2 / [\exp(\Omega/k_B T) - 1]$ [3, 4], in which the linear term accounts for acoustic phonons and the last term for optical phonons of mean energy Ω . The fit gives $\Gamma_0 = 5.5 \pm 1.0$ meV and $\Omega = 46 \pm 7$ meV, close to the E_{2g}^1 and A_{1g} phonon energies of WS_2 (~ 44 and ~ 51 meV), while $c_1 = 9 \pm 15$ $\mu\text{eV/K}$ is consistent with zero. Above ~ 250 K the optical-phonon occupation grows nearly linearly with temperature, so the linewidth is well approximated by a straight line. A linear fit over 250–1000 K gives 0.129 ± 0.002 meV/K

with an rms residual of 1.7 meV, slightly smaller than that of the model over the same range (1.9 meV); this linear relation is the thermometric indicator used in the main text. Above ~ 1100 K the width levels off as the low-energy tail of the emission extends beyond the acquisition window (Fig. S4), and these points are excluded from both fits. The decomposition of the Voigt profile (Eq. 3, Sec. S4) into its Gaussian (Γ_G) and Lorentzian (Γ_L) contributions is not reported here: the two components are strongly correlated in the fit and are not individually well determined over much of the temperature range.

Figure S5(b) shows the integrated PL intensity of X_A^0 . The emission is weak below ~ 40 K and then rises by roughly two orders of magnitude up to ~ 175 K. This behaviour is characteristic of tungsten-based monolayer TMDs, in which the lowest-lying exciton state is spin-forbidden (dark) and sits a few tens of meV below the bright state [5, 6]: at low temperature carriers relax into the dark state and bright emission is quenched, while increasing temperature thermally activates the bright-state population. Above ~ 175 K the intensity decreases again, consistent with thermally activated non-radiative recombination [7] and with the thermal escape of excitons from the radiative window [8]. Above ~ 800 K the extracted intensity is no longer a faithful measure of the emission strength, since the strongly broadened peak extends beyond the low-energy edge of the fitting window (1.50 eV) and its integrated area cannot be reliably determined. The peak *position* is unaffected by this truncation, since the emission maximum remains well inside the fitting window at all temperatures.

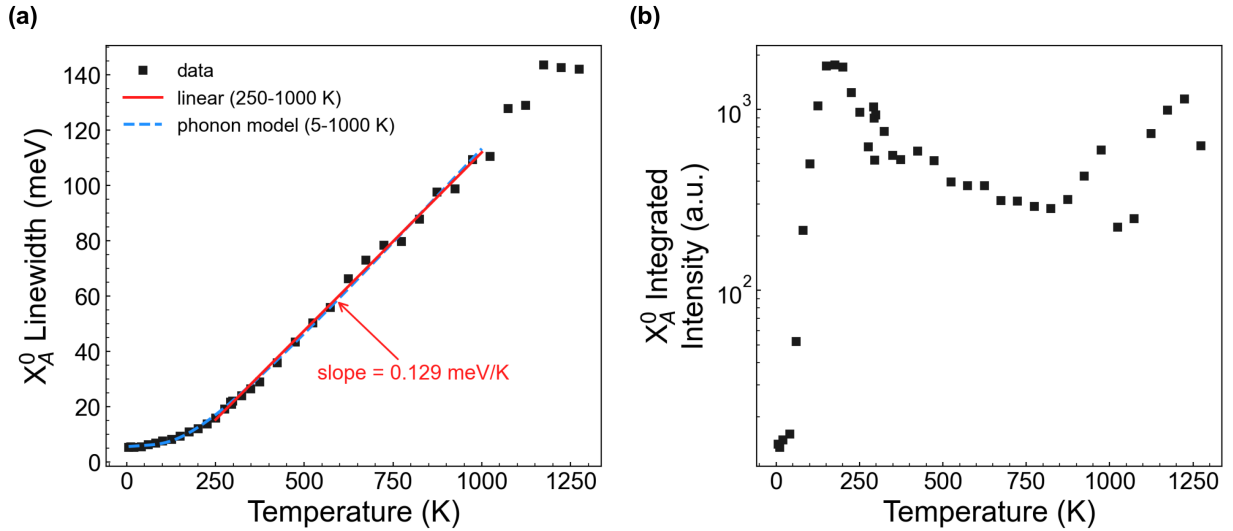


Figure S5: Temperature dependence of the PL linewidth and intensity. (a) Total Voigt FWHM Γ_V of the X_A^0 emission versus temperature. Blue dashed line: phonon-scattering model fitted over 5–1000 K; red line: linear fit over 250–1000 K. (b) Integrated X_A^0 PL intensity versus temperature, showing the thermally activated rise from the dark-exciton ground state below ~ 175 K and the subsequent thermal quenching at higher temperature.

S6. Thermal Dissociation of WS_2 Monolayer

Figure S6 shows optical images of the non-encapsulated sample on the SiC membrane before and after thermal treatment at 1000°C . After this treatment no WS_2 Raman signal was detectable, and the monolayer is absent from the membrane under white-light illumination, consistent with thermal dissociation through sulfur sublimation [9]. The hBN-encapsulated sample, by contrast, remained intact after the same treatment, with both Raman and PL still measurable at 1273 K. Thermal dissociation therefore sets the practical upper bound for optical thermometry on bare monolayers at ~ 1273 K, while the encapsulated heterostructure was limited here only by the operating range of the heater chip.

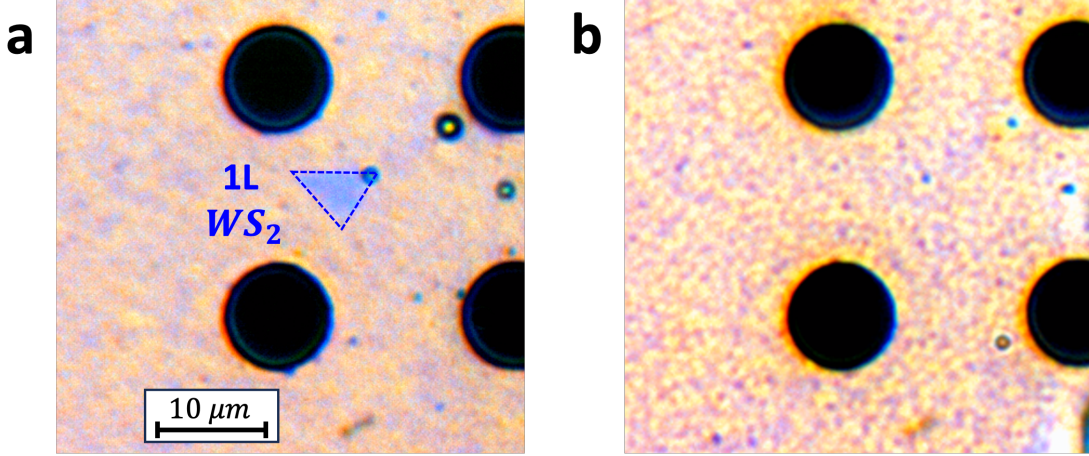


Figure S6: Thermal dissociation of monolayer WS_2 . Optical images of the SiC membrane (a) before annealing, showing an exfoliated WS_2 monolayer flake (dashed outline), and (b) after annealing to 1000 °C. The WS_2 monolayer is no longer visible, confirming thermal dissociation near 1273 K through loss of sulfur species.

S7. Raman Intensity and Excitonic Resonances

Figure S7 shows the integrated intensity of the $\text{E}_{2g}^1 + 2\text{LA}$ band and the A_{1g} mode versus temperature (arbitrary units, not comparable between the two samples). The temperature dependence reflects resonance Raman scattering [10], which is strongest for the double-resonant 2LA(M) band [11]. As the excitonic transitions redshift with temperature, the B exciton (~ 2.4 eV at room temperature [12]) crosses the excitation energy (2.410 eV), giving the sharp maximum near 225 K in the encapsulated sample and the maximum at 298 K in the non-encapsulated one. The recovery of both bands above ~ 900 K in the encapsulated sample is attributed to the approach of the higher-lying C exciton resonance (~ 2.8 eV at room temperature [12]), a regime that the non-encapsulated monolayer does not reach because it dissociates above ~ 1223 K (Sec. S6).

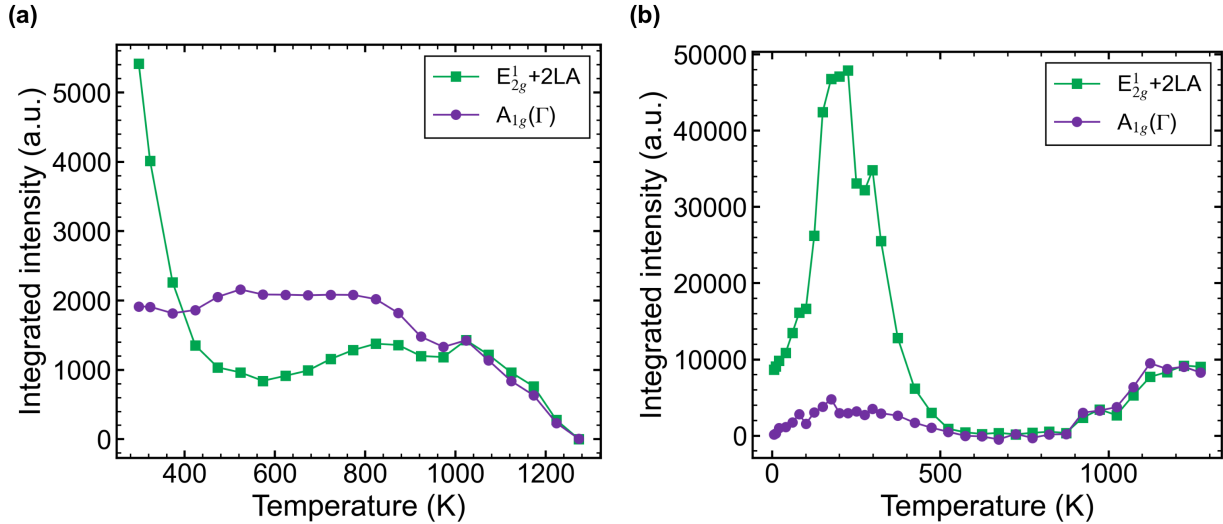


Figure S7: Integrated Raman intensity of the $\text{E}_{2g}^1 + 2\text{LA}$ band (green squares) and the A_{1g} mode (purple circles) versus temperature for (a) the non-encapsulated and (b) the hBN-encapsulated sample. Intensities are in arbitrary units and are not comparable between the two samples.

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